Chapter-8
Fundamentals of thermodynamics and heat transfer

8.3. Heat transfer:

Heat is a common form of energy which is continuously being transferred from one body to another. In earlier time, during the development of concept of heat and temperature, people had only the concept of inequality for flow of heat. According to this concept, heat always flows from a body of higher temp. to a body of lower temp. Actually, heat always flows until there is difference in temp.

There are three different modes of transfer of heat. Transfer of heat due to actual motion of molecules in fluid (liquid and gas) is called Convection, eg. boiling of water. Transfer of heat due to molecular vibrations in a solid is called Conduction eg. heating of metal rod. Transfer of heat without support of any material medium is called radiation, eg. receiving heat from sun.

These modes of heat transfer are described as following:

(1) Conduction:

Conduction is the process of transfer of heat from one point to another point of a body carried out by means of collisions between rapidly vibrating atoms at hotter region and slowly vibrating atoms at colder region. There is no actual transfer of particles during conduction.

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when one end of a solid is heated, atoms at hotter end vibrate with greater amplitude and have more K.E. than neighbouring atoms at colder part. The atoms at hotter region collide with and give up some energy to the neighbouring atoms in colder regions. Similar process continues between each set of neighbouring atoms up to next end. As a result transfer of heat takes place and previously cold end also gets heated. This method of transfer of heat is called conduction.

(2) Convection:

Convection is the process of ~~heat~~ transfer of heat in fluids by means of actual motion of heated particles from higher temperature region to lower temp. region.

Heated particles carry heat and move from hot region to cold region but cold particles move in opposite direction. The current set up in the process is called convection current. This method is not possible in solid and vacuum. Heating of water, land breezes, sea breezes, wind etc. are some examples of convection.

(3) Radiation:

Radiation is the process of transmission of heat from one point to another without need of any material medium. Radiation does not heat the medium through which heat energy passes

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In radiation, heat is transferred in the form of e.m. radiation (wave) and travel with speed of 3×10^8 m/sec. in vacuum. Heat energy coming to the earth from the sun, and transmission of heat around fire are the examples of radiation. Heat energy radiated by an object is called radiant energy or thermal energy.

(II) Statement and assumption of Fourier law of thermal conductivity:

The ability of anybody to conduct heat is measured in term of thermal conductivity. Therefore, Ability which measures the thermal conduction of material is called thermal conductivity. Thermal conductivity generally occurs in solids.

Statement of Fourier law of Thermal conductivity:

The Fourier law states that the rate of heat flow in solids is directly proportional to the cross-section area perpendicular to the flow axis and negative of temp. gradient over the length of the path of conduction.

According to Fourier's law, the rate of heat flow Q through a homogenous solid is directly proportional to the area A of the section at the right angles to the direction of heat flow, and to the temp. difference dT along the path of heat flow.

i.e. $$Q = -KA\frac{dT}{dx} \quad \text{---(1)}$$

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Where

Q = rate of heat transfer (watt)

K = Thermal Conductivity ($W/m \cdot K$)

A = Area of cross-section (m^2)

dT = Change in temp. along the dirn. of heat flow

dx = Thickness of the object.

In the fourier heat conduction eqn. the -ve sign implies that heat is flowing from higher temp. to lower temp. therefore it is provided to compensate for the negative nature of the temp. gradient.

Assumption in fourier law of heat conduction:

Following are the assumptions of fourier law of heat conduction:

1. The thermal conductivity of the material is constant throughout the material.
2. There is no internal heat generation that occurs in the body.
3. The temp. gradient is considered at constant.
4. The heat flow is unidirectional and takes place under steady-state conditions.
5. The surfaces are isothermal.

Temperature gradient

The rate of fall in temp. with distance along the dirn of heat flow is called the temp. gradient.

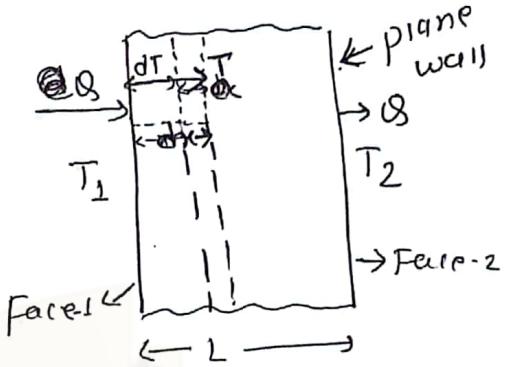
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It is expressed as $-\frac{dT}{dx}$ and its unit is kelvin per meter (K/m). Here dT is the small change in temp. over a small distance dx and -ve sign indicates that temp. falls with distance.

One dim^n steady state heat conduction through plane wall

Consider parallel sided plane wall of thickness L and uniform cross-section area A. as shown in figure. let K be the thermal conductivity of wall material through which heat is flowing only in x-direction.



[Fig: plane wall]

Let T_1 and T_2 are the temp. of higher temp. face-1 and lower temp. face-2 The small change of temp. dT when the particles conduct every small distance dx.

According to fourier law, the rate of heat transfer is given by

$$\frac{dQ}{dt} = -KA\frac{dT}{dx}$$

$$\text{or } \frac{dQ}{dt}dx = -KA\,dT \quad \text{---(1)}$$

To obtain the rate of flow of heat through this whole wall is obtained by integrating eq^n (1), we get

$$\int_0^L \frac{dQ}{dt}dx = \int_{T_1}^{T_2} -KA\,dT$$

$$\text{or } \frac{dQ}{dt}\int_0^L dx = -KA\int_{T_1}^{T_2} dT$$

$$\text{or } \frac{dQ}{dt}[L-0] = -KA[T_2-T_1]$$

$$\text{or } dQ/dt = \frac{KA(T_1-T_2)}{L}$$

$$\therefore \frac{dQ}{dt} = \frac{(T_1-T_2)}{L/KA} \quad \text{---(2)}$$

where $\frac{L}{KA}$ = Thermal resistance R_{th}

$$\therefore \frac{dQ}{dt} = \frac{T_1-T_2}{R_{th}} \quad \text{---(3)}$$

This eq^n (3) gives the rate of flow of heat through the wall. If R_{th} increases, $\frac{dQ}{dt}$ decreases and vice-versa.

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Black body and black body radiation

A perfect black body is one which absorbs heat radiation of all wavelengths falling on it and emits radiations of all possible wavelengths when heated.

For a perfect black body absorptance (a) = 1 and emissivity (e) = 1. When a body absorbs all radiant energy, it neither reflect nor it transmit. As a result body appears black and known as black body. Sun emits radiations of all wavelengths. So it can be considered as a black body. A good absorber is a good emitter. Therefore, a black body is a good emitter and can emit the radiation of all possible wavelengths

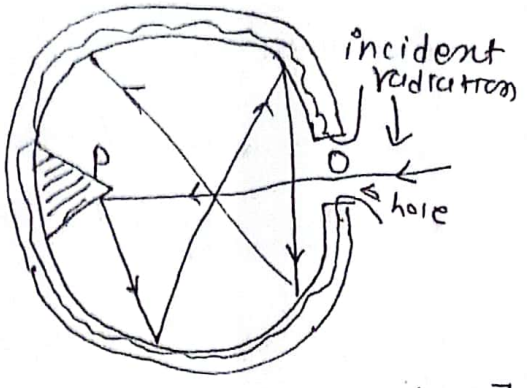
A perfect black body cannot be realized in actual practice. The nearest approach to a perfectly black body is a surface coated with lamp black or platinum black. Such a surface absorbs 96% to 98% of the incident radiation. Ferry's body is an example of perfectly black body realized in practice.

Ferry's black body:

Ferry's black body is a close approximation of perfectly black body which can be realized in practice. Ferry's body consists of a double walled hollow copper sphere with a small fine hole O and a conical projection P opposite to O as shown in figure.

A heating filament is kept in between two ~~boats~~ walls of the sphere and inner surface of sphere is coated with lamp black or platinum black

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[Fig: Ferry's black body]

When heat radiation falls on the conical projection P through hole O, projection P prevents the direct reflection of incident radiation rather it causes multiple reflections of radiations inside the sphere. most of the incident radiation is absorbed by lamp black or platinum black and remaining part is also lost by multiple reflection. When the sphere is heated by passing electric current to the heating filament, radiations of all possible wavelengths come out from the hole O. Hence, the hole O behaves as a black body.

Emissive power

It is the amount of heat radiations radiated by a unit area of a black body per unit time at a particular temp. It is denoted by E. Mathematically,

$$\text{Emissive power } E = \frac{\text{Energy radiated } (Q)}{\text{Area } (A) \times \text{time } (t)} = \frac{Q}{A \times t}$$

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$$\therefore E = \frac{Q/t}{A} = \frac{P}{A}$$

$$\text{unit of } E = \frac{J/sec}{m^2} = W/m^2.$$

According to Stefan's-Boltzmann law,

$E = \sigma T^4$ where σ = Stefan-Boltzmann constant $= 5.67 \times 10^{-8}\ W/m^2 K^4$.

Emissivity (e):

Emissivity of a body is defined as the ratio of emissive power of a given body to the emissive power of a perfectly black body of same size at the same temp. It is denoted by e.

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$$\therefore e = \frac{\text{emissive power of body}}{\text{emissive power of perfect black body.}} = \frac{E}{E_b}$$ of

Emissivity has no unit and dim^n Value of e depends upon the nature of the surface of the emitting body. It values ranges from 0 to 1.

For a perfectly black body, $e = 1$

" a black body $e < 1$

For a perfectly reflecting surface $e = 0$.

Stefan's-Boltzmann's law of blackbody radiation:

Stefan's declared experimentally that energy radiated by a body depends only on its temp. and he put forward a law known as Stefan's law.

Stefan's-Boltzmann law states that " emissive power of a black body is directly proportional to the fourth power of absolute temp. of surface of the body".

let E be the emissive power of a black body at absolute temp. T. Then, according to Stefan's-Boltzmann law $E \propto T^4$

or $E = \sigma T^4$ — (1) where σ is proportionality constant is called stefan's Const. The value of σ in S.I. system is $5.67 \times 10^{-8}\ W m^{-2} K^{-4}$.

If a body is not perfectly black, and has its emissivity e. Then,

$E = e\sigma T^4$ — (2)

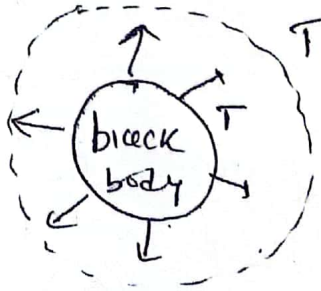


Fig: Stefan's law of

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or $\frac{Q}{A\,t} = e\sigma T^4$ or $\frac{Q/t}{A} = e\sigma T^4 \Rightarrow \frac{\text{power}}{A} = e\sigma T^4$

$\therefore P = eA\sigma T^4$ —(3)

The relations given above are valid only if the temp. of the surrounding is zero kelvin. But, if there is a surrounding at temp T_0 (more than 0K.) then, surrounding also emits heat radiation, which is absorbed by black body. Hence, relation is modified by Boltzmann for the net emissive power as

$E' = \sigma(T^4 - T_0^4)$ —(4) This is called Stefan's-Boltzmann law.

If a body is not perfect black body, the net emissive power $E' = e\sigma(T^4 - T_0^4)$

$\frac{Q'}{A\,t} = e\sigma(T^4 - T_0^4)$

or $\frac{Q'/t}{A} = e\sigma(T^4 - T_0^4)$

$\therefore P' = eA\sigma(T^4 - T_0^4)$ —(5)

This gives power of radiated energy of a black body.

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Some Numerical problems

① An insulating material having a thermal conductivity of 0.08 W/m-K is used to limit the heat transfer of 80 W/m² for a temp. of 15°C across the opposite faces. Find the thickness of the material.

Sol: $K = 0.08$ W/m-K, $\frac{Q}{tA} = 80$ W/m², $\Delta T = 15°C = 15K$

$dx = ?$ $\Rightarrow \frac{Q}{t} = KA\frac{dT}{dx} \Rightarrow dx = \frac{Q}{tA}\frac{dT}{K} \Rightarrow dx = 0.15m$

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(2) A brick wall 12 cm thick and $5m^2$ surface area exposed to 50°C at one face and 20°C to another face. If the thermal conductivity of the material is 1.5 W/m-K. Find the heat transfer rate.

Soln: $\Delta x = 12cm$, $A = 5m^2$ $dT = 50°C - 20°C = 30°C = 30K$.

$K = 1.5 W/m-K$, $\frac{Q}{t} = ?$ $\frac{Q}{t} = KA\frac{dT}{dx} = 1875$ watt.

(3) Find the rate of heat lost from a brick wall ($k = 0.7 W/m-K$) of length 5m, height 4m and 0.25m thick. The temp. of the inner surface 50°C and that of outer surface is 30°C. Also, Calculate the distance from the inner surface at which the temp. is 40°C

Soln: k of brick wall = 0.7 W/m-K.

Area $A = 5m \times 4m = 20m^2$,

$dx = 0.25m$, $dT = 50°C - 30°C = 20°C = 20K$

$\frac{dQ}{dt} = KA\frac{dT}{dx}$ = 1120 watt

Again, $dx' = ?$ where $\frac{dQ}{dt} = \frac{KA(T_1 - T_2)}{dx'}$ $\Rightarrow 1120 = \frac{0.7 \times 20(50-40)}{dx'} \Rightarrow 0.125$ m

(4) A 1.2m long tube with outer diameter of 4cm outside temp of 120°C is exposed to an ambient air at 20°C. If the heat transfer coefficient between the tube surface and the air is 20 W/m²K. Find the rate of heat transfer from the tube to the air.

Soln: $dx = 1.2m$

$dT = 120°C - 20°C = 100°C = 100K$, Heat transfer rate $h = \frac{K}{dx} = 2000 W/m^2-K$.

$A = 2\pi R dx = 2\pi \times 2 \times 10^{-2} \times 12 = 0.1508 m^2$.

$\frac{dQ}{dt} = \frac{K}{dx} dT \cdot A = 301.593$ watt

(5) A furnace inside temp of 2250K has a glass circular viewing of 6cm diameter. If the transmissivity of glass is 0.08 make calculations for the heat loss from the glass window due to radiation.

Soln: $A = \frac{\pi}{4}d^2 = \frac{\pi}{4}(6 \times 10^{-2})^2 =$ $e = 0.08$ $T = 2250K$ → $E = \sigma T^4 \Rightarrow \frac{Q}{t \cdot A} = \sigma T^4 \times e$ for not perfect

$\Rightarrow \frac{Q}{t} = \sigma T^4 e \cdot A = 328.59$ watt.

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Chapter-8

8.1 Fundamentals of thermodynamics

Phy-1

Introduction:

When two stones strike together spark may be produced. ~~When two dry woods are rubbed together fire may be~~ When we rub our hands each other, we feel warm. In both cases heat is produced in the expense of mechanical ~~work~~ energy. Also, when certain mass of ice falls from a height, it may melt. This is due to conversion of gravitational potential energy into heat energy. We have other examples such as in cars, bikes, airplanes etc. fuel is burnt to produce heat which causes the expansion of gases inside the cylinder and hence mechanical work is obtained. In all cases, we mentioned the principle of conservation of energy is not violated. Transformation of heat into mechanical work and vice-versa has provided us the ~~found~~ foundation for the construction of heat engines (like bike, scooter, cars etc) and refrigerators. Thermodynamics is the branch of physics which deals with the interrelationship between heat and mechanical work. Now a days thermodynamics has been more applicable in other fields like electrical, chemical and magnetic fields.

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Note:

Thermodynamics ⇒ Thermo + dynamics = Heat + Study of motion.
It is the branch of physics which deals with the transformation of heat energy into mechanical work and vice-versa. Thermodynamics deals macroscopically i.e. it deals with the bulk system but not with the individual molecule.

Thermodynamic system (Types of thermodynamic system) & its typ

An assembly of large number of particles bounded by a closed surface (boundary surface) having certain value of

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macroscopic parameters (pressure, volume and Temp) of concern. Thermodynamic system. Anything outside the boundary surface of system is called its surrounding. Thermodynamic system interacts with its surrounding either by exchanging energy and matter in different ways or by performing work.

On the basis of exchange of energy (heat) and matter, systems are classified into three types. They are:

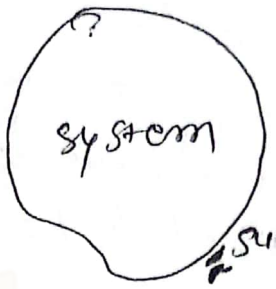


fig: Thermodynamic system

(1) Open system: Thermodynamic system which can exchange both matter and energy with its surrounding is known as open system. eg. gas contained in an open cylinder.

(2) Closed system: Thermodynamic system which can exchange only energy but not matter with its surroundings is called closed system. eg. gas contained in a ~~perfectly~~ closed and conducting cylinder fitted with piston.

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(3) Isolated system: Thermodynamic system which can exchange neither energy nor matter with its surrounding is called Isolated system. eg. gas contained in a perfectly insulated cylinder fitted with piston.

Thermodynamical Variables and properties

An equilibrium state of a thermodynamic system is specified by the value of its parameters like pressure(p), Volume (V), and Temp (T) etc. These parameters are called Thermodynamic Variables.

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On the other hand, thermodynamic property is any characteristics of thermodynamics variable and other additional terms like mass, internal energy, total energy, enthalpy, entropy, viscosity, thermal conductivity etc.

Thermodynamic properties are generally classified into two groups. They are:

(1) Intensive properties: The properties which are independent of the mass of the system are called intensive properties. eg. Temp. pressure, ~~density~~ etc. are examples.

(2) Extensive properties: The properties which depend on the size or extent of the system are called extensive properties. eg. Total mass, total volume and ~~total~~ momentum

(II) Thermodynamic equilibrium:

Thermodynamic deals with equilibrium states. Equilibrium means a state of balance. An equ^m state, there is no existed unbalanced potentials within a system. A system will be in a state of thermal equ^m, if the condition for the following three types of equ^m are satisfied.

(1) mechanical equ^m: A system is in mechanical equ^m if there is no change in pressure at any point of the system with time.

(2) Thermal equ^m: A system is said to be in thermal equ^m if all parts of it exhibit the same value of temp.

(3) Chemical equ^m: A system is in Chemical equ^m if its Chemical composition does not change with time, that is no chemical reaction occur.

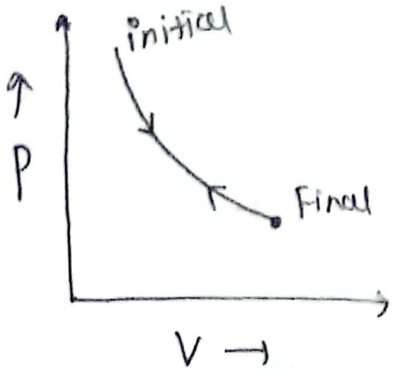
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Thermodynamic process (Reversible and Irreversible process)

(a) **Reversible process:**

If the effects produced by a process on the system as well as on the surrounding can be completely restored to their initial states and no changes are left in any of the system taking part in the process or in the surrounding then the process is said to be a reversible process. Any process, taking place slowly and controlled manner. All the mechanical process taking place under the action of conservative force are reversible process. In reversible process, a system can retrace in the opposite dir^{n} so that it passes through exactly the same states in all respect as in the direct process.



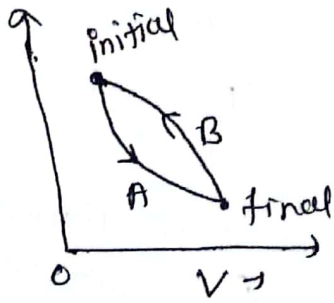
[Fig: Reversible process]

Conditions for the reversible process

1. The process should be infinitely slow
2. There should be no loss of energy during the process.
3. System and surrounding should be in thermodynamic equ^{m} during the process.
4. No dissipative force of friction, viscosity, electrical resistance etc should be presented.

(b) **Irreversible process**

The process which produce a permanent change in the thermodynamic state of the system and cannot be retraced in the opposite order are called irreversible process. Almost all processes are in the nature of irreversible process.



[Fig: irreversible process]

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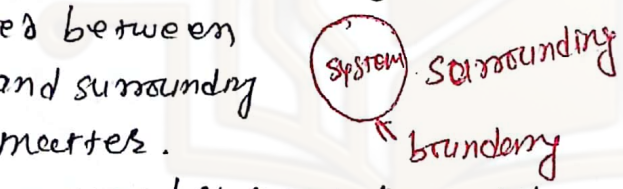
Eg. rusting of iron, decay of matter, heat produced by friction etc. are irreversible process.

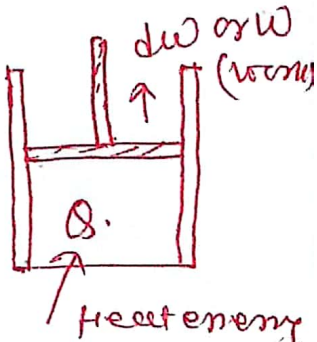
8.2 Laws of thermodynamics

First law of thermodynamics is based on the principle of conservation. It states that "energy can neither be created nor can be destroyed but can change one form to another form". This law can be expressed in closed and open systems. For a closed system 1st law of thermodynamics can be explained based on the energy conservation. In open system, it can be explained based on the mass conservation and energy conservation. They are described as following:

First law of thermodynamics for closed system (For control mass):

In the closed system, energy can be transferred between system and surrounding but not matter.





Consider a closed system, amount of heat energy Q is given as an input and external work done (W) as an output as shown in figure.

Using the law of conservation of energy.

Energy entering the system − Energy leaving the system = Change in total energy of the system.

i.e. $E_{in} - E_{out} = \Delta E$

$$Q - W = \Delta E. \quad \text{—(1)}$$

The total energy of the system (i.e. change of energy is the sum of K.E., P.E and internal energy.)

$\therefore \Delta E = \Delta K.E + \Delta P.E + \Delta U$ where ΔU = Change in internal energy of the system. (2)

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In the case of closed system, change in KE and PE are negligible as compared to change in internal energy. Therefore, from eqn (2), we get

$$\Delta E = \Delta U \quad \text{— (3)}$$

Using eqn (3) in eqn (1) we get

$$Q - W = \Delta U$$

$$\text{or } dQ - dW = dU$$

$$\text{or } dQ = dU + dW \quad \text{— (4)}$$

This is an eqn. of 1st law of thermodynamics.

Case-1: If the closed system undergoes in cyclic process:

The conservation of energy eqn can be written as,

$$\oint (dQ - dW) = \oint dU$$

For cyclic process $\oint dU = 0$

$$\therefore \oint (dQ - dW) = 0$$

$$\text{or } \oint dQ = \oint dW$$

Therefore, net amount of applied heat energy (Q) is equal to net amount of external work done (W).

Case-II: For any thermodynamical process from one to another ~~systems~~ of a closed systems.

$$Q_{12} = W_{12} = U_2 - U_1$$

$$Q_{12} = W_{12} + U_2 - U_1.$$

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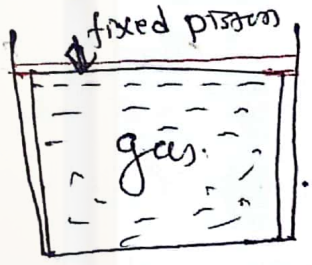
Application of 1st law of thermodynamic of a Closed System

Some appeication of 1st law of thermodynamics are described as following:

① Constant Volume (Isochoric) process:

It is a thermodynamic process in which vol^m of gas remains constant as the gas is kept in a non-expanding chamber.

Consider the working substance (gas) is contained in a rigid vessel in a constant volume process, making the system's boundaries immovable and preventing any work from being done on or by the system. The rigid vessel as shown in figure.



[Rigid vessel]

when the heat energy is added in the system, there is no change of vol^m. so, $dV = V_2 - V_1 = 0$

Total work done $dw = pdV = 0$

We know that, from 1st law of thermodynamics,

$$Q_{12} = W_{12} + U_2 - U_1$$

$$Q_{12} = U_2 - U_1 \quad \text{Since } dw = W_{12} = 0$$

Thus, $Q_{12} = (\Delta U)_{12}$

Hence, heat transferred to control mass during a constant volume process is equal to the increase in internal energy of the system.

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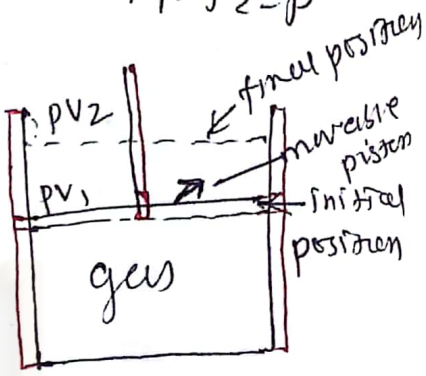
(2) Isobaric process:

It is a thermodynamics process in which the working substance is taken in an expanding chamber and heat is supplied to it so that volume and temp. of system change without any change in pressure.

In the Isobaric process, work done

$$\int dw = w_{12} = \int_{v_1}^{v_2} p\,dv = p(v_2 - v_1) \quad \text{where } p_1 = p_2 = p$$

$$\therefore w_{12} = pv_2 - pv_1 \quad \text{---(1)}$$



From 1st law of thermodynamics,

$$Q_{12} = w_{12} + u_2 - u_1$$

or $Q_{12} = pv_2 - pv_1 + u_2 - u_1$

or $Q_{12} = (u_2 + pv_2) - (u_1 + pv_1)$ ---(2)

or $Q_{12} = H_2 - H_1$ where $H_1 = u_1 + pv_1$ and $H_2 = u_2 + pv_2$

Since $H = U + pv$ is called enthalpy is equal to the sum of internal energy and product of pressure and volume of the gas.

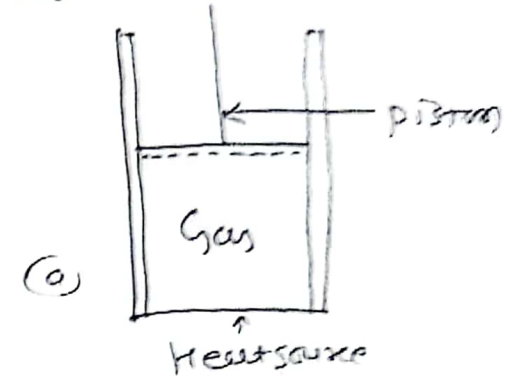
From eq(2), we conclude that heat transferred to a control mass during a const. pressure process is equal to the increase in the enthalpy of the system.

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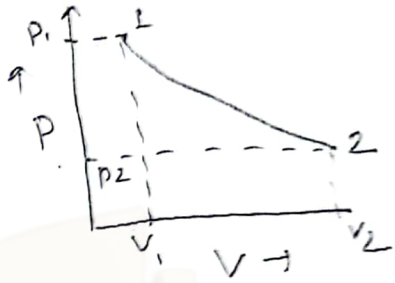
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3 Isothermal process

It is an isothermal process in which temp of a system remains constant throughout the process. There is free exchange of heat between system and surrounding. So it is a closed system.



Consider an ideal gas is confined in the conducting cylinder as shown in figure. let p, V and T are pressure, volume and Temp of a gas. In isothermal process temp. remain constant. Therefore.



eqⁿ of 1st law of thermodynamics

$$dQ = dU + dW$$

or $$Q_{12} = W_{12} + U_2 - U_1 \quad \text{—(1)}$$

We know that internal energy is a function of temp.

i.e. $U_2 - U_1 = dU = C_V dT = 0$ since at const temp $T_2 - T_1 = dT = 0$

$\therefore$ eqⁿ (1) will be

$$Q_{12} = W_{12} + 0 \Rightarrow Q_{12} = W_{12} \quad \text{—(2)}$$

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To find the work done during an isothermal process, we consider an ideal gas filled in the conducting cylinder fitted with piston.

The amount of work done during isothermal process

$$dW = p\,dV \Rightarrow \int dW = \int_{V_1}^{V_2} p\,dV$$

or $$W = \int_{V_1}^{V_2} p\,dV \quad \text{—(3)}$$ since in the process of work done, vol^m change V_1 to V_2 by taking const Temp.

We know that $pV = nRT = \text{const.}\ (k)$

$$\therefore p = \frac{C}{V}$$

Then eqⁿ (3) will be.

$$W = \int_{V_1}^{V_2} \frac{C\,dV}{V} \Rightarrow W = C\left[\log_e V\right]_{V_1}^{V_2} \Rightarrow W = C \log_e\left(\frac{V_2}{V_1}\right)$$

—(4)

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we know that for isothermal process

$p_1V_1 = p_2V_2 = \text{Const}\ (C.)$ Thus, using eq^n (5) in eq^n (4) $w =$

$\therefore Q_{12} = W_{12} = p_1V_1 \ln\left[\frac{V_2}{V_1}\right]$ or

or

$Q_{12} = W_{12} = p_2V_2 \ln\left(\frac{V_2}{V_1}\right)$

Hence heat transferred to a control mass for an ideal gas during a constant temp. process is equal to work done by the gas

(a) Adiabatic (polytropic) process

It is an thermodynamic process in which pressure, volume and temp. of the system change without exchange of heat between system and surrounding. Therefore, from 1st law of thermodynamics,

$dQ = du + dw$ or $Q_{12} = W_{12} + U_2 - U_1$ —(1)

~~$\Rightarrow du = dw$ where $dQ = 0$~~

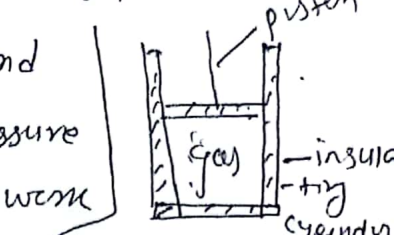
~~Also, we noted that~~

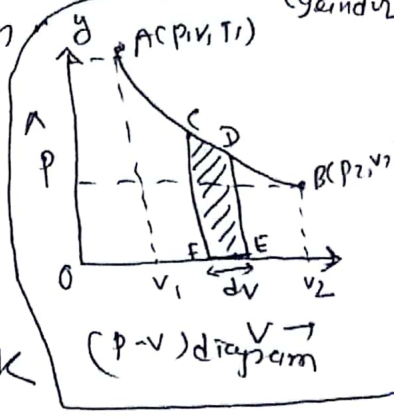
~~$Q_{12} = W_{12} + U_2 - U_1$~~

~~$\downarrow 0$~~

~~$\therefore -W_{12} = U_1 - U_2$~~

Suppose an n moles of an ideal gas taken in a insulating cylinder as shown in figure The cylinder has perfectly insulating walls and fitted with frictionless, massless non-conducting piston. In this case, there will be no exchange of heat in between system and surrounding. Suppose gas expand at (p_1, V_1, T_1) to (p_2, V_2, T_2). If p is the pressure of the gas at an instant ~~of time~~ then the work done for small change in vol^m dv is given by $dw = pdv$. The total work done ~~at~~ during whole process is





$$W = \int_{V_1}^{V_2} pdv \quad —(2)$$

For an adiabatic process $pV^\gamma = \text{const.} = K$

$$\therefore p = \frac{K}{V^\gamma} \quad —(3)$$

Using eq^n (3) in eq^n (2) we get

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(3) $W = \int_{V_1}^{V_2} \frac{k\,dv}{V^\gamma}$

or $W = k \int_{V_1}^{V_2} V^{-\gamma} dv$

or $W = k \left[\frac{V^{-\gamma+1}}{-\gamma+1}\right]_{V_1}^{V_2}$

or $W = \frac{k}{1-\gamma}\left[V_2^{1-\gamma} - V_1^{1-\gamma}\right]$

or $W = \frac{1}{(1-\gamma)}\left(kV_2^{1-\gamma} - kV_1^{1-\gamma}\right]$ —④

we know that for adiabatic process

$p_1V_1^\gamma = p_2V_2^\gamma = k$ so eqⁿ ④ will be

$W = \frac{1}{(1-\gamma)}\left[p_2V_2^\gamma \cdot V_2^{1-\gamma} - p_1V_1^\gamma \cdot V_1^{1-\gamma}\right]$

or $W = \frac{1}{(1-\gamma)}\left(P_2V_2 - P_1V_1\right]$ —⑤

→ for ideal eqⁿ of gas $P_1V_1 = nRT_1$

$P_2V_2 = nRT_2$

So, eqⁿ ⑤ will be

$W = \frac{1}{(1-\gamma)}(nRT_2 - nRT_1)$

$\therefore W = \frac{nR}{(1-\gamma)}(T_2 - T_1)$ —⑥

where n = mole of gas.

Using eqⁿ ⑥ in eqⁿ ① we get

$Q_{12} = \frac{nR}{(1-\gamma)}(T_2 - T_1) + nC_v(T_2 - T_1)$

where $du = nC_v(T_2 - T_1)$ or nC_vdT.

and C_v = molar sp. heat capacity of a gas at constant volume.

↳ $\therefore Q_{12} = n\left[\frac{R}{1-\gamma} + C_v\right]T_2 - T_1$

or $Q_{12} = nC_n(T_2 - T_1)$ where $C_n = \left(\frac{R}{1-\gamma} + C_v\right)$ is called polytropic specific heat capacity of mole of gas.

(5) Specific heat capacity:

⇒ Specific heat of a gas is defined in terms of standard mass system (kg), it is called principal specific heat capacity (c)

⇒ If specific heat of gas defined in terms of molar mass, is called molar specific heat capacity (C)

Specific heat capacity of a gas is explained by two ways i.e. taking volume constant (C_v) and pressure constant (C_p) They are described as following:

(1) Sp. heat capacity at constant volume:

(a) principal Sp. heat capacity of gas at constant volume (C_v):

It is defined as the quantity of heat required to raise the temp. of 1 kg of a gas through 1 kelvin temp at const. volume

In SI unit C_v is $[kJ\,kg^{-1}K^{-1}]$

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(b) molar specific heat capacity at constant volume (C_v):

C_v is defined as the amount of heat required to raise the temp of one mole of gas through one kelvin temp at const. volume. Its SI unit $J\,mol^{-1}K^{-1}$. M be the molar mass of gas in kg. Then

$C_v = mc_v$.

specific heat of an ideal gas at constant volume can be defined in term of change in internal energy. So, internal energy of an ideal gas can be determined if its temp. and specific volume are given.

$\therefore u = u(T, v)$

$$du = \left(\frac{du}{dT}\right)_v dT + \left(\frac{du}{dv}\right)_T dv$$

For constant volume, $dv = 0$

$$\therefore du = \left(\frac{du}{dT}\right)_v dT$$

or $du = C_v\,dT$ where $C_v = \left(\frac{du}{dT}\right)_v$.

For n mole of gas, $du = n\,C_v\,dT$

If m mass of gas $du = mC_v\,dT$

∴ Specific heat at constant volume is defined as a change in specific internal energy per degree change in temp.

(2) Specific heat capacity at constant pressure

(a) principal specific heat capacity at constant pressure (C_p):

C_p is defined as the quantity of heat required to raise the temp. of 1 kg of gas through 1 K at constant pressure. In SI unit, C_v has $J\,kg^{-1}K^{-1}$.

(b) molar specific heat capacity at constant pressure (C_p):

C_p is defined as the amount of heat required to raise the temp. of 1 mole of gas through 1 K at constant pressure. If ~~kets~~ units is $J\,mol^{-1}K^{-1}$. If M be the molar mass of a gas in kg. Then $C_p = MC_p$.

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The sp heat capacity of an ideal gas at constant pressure can be defined in terms of change in enthalpy so we have, specific enthalpy

$$H = H(T, P)$$

Change in sp enthalpy is given as

$$dH = \left(\frac{dH}{dT}\right)_P dT + \left(\frac{dH}{dP}\right)_T dP$$

For constant pressure $dP = 0$

$$\therefore dH = \left(\frac{dH}{dT}\right)_P dT$$

or $dH = C_p dT$ where $C_p = \left(\frac{dH}{dT}\right)_P$ is called sp. heat at constant pressure and is defined as the change in sp. ~~heat~~ enthalpy per degree change in temp.

H.W Show that $C_p > C_v$ [from +2 book]

② Derive the relation $C_p - C_v = R$] [from +2 book]

Second law of thermodynamics

limitations of first law of thermodynamics

First law of thermodynamics is an important tool for understanding physical processes. Also no any process can violate it but has four limitations. They are given as following:

① When two bodies at different temp. are brought in thermal Contact, heat flows always from hot body to cold body but first law of thermodynamics is silent to the question why heat doesn't flow from cold body to hot body. Therefore, first law of thermodynamics does not indicates the direction of heat flow.

② Heat can not be converted into useful work from a

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The sp. heat capacity of an ideal gas at constant pressure can be defined in terms of change in enthalpy. So we have, specific enthalpy

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H.W Show that $C_p > C_v$ [from +2 book]

Ans (1)

(2) Derive the relation $C_p - C_v = R$] [from +2 book]

Second law of thermodynamics

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Single heat reservoir like sea. But why this is impossi
First law of thermodynamics has no answer to this que
That is first law of thermodynamics does not provide the
Condition under which heat can be converted into wor

(3) When a bullet strikes a target, K.E. of bullet converts into heat energy but target cannot provides energy to the bullet for fly back losing its heat why? First law of thermodynamic has no answer that whether the mechanical work and heat energy are inter convertible or not.

(4) In a heat engine, allowed by first law of thermodynamic, no 100% heat energy converts into mechanical works, why? First law of thermodynamics has no answer to this. That is first law of thermodynamics does not indicate that to what extent, heat energy can be converted into mechanical work.

Second law of thermodynamics

Second law of thermodynamics deals with the nature or quality of ~~heat~~ energy and defines the direction in which the systems can proceed. Hence, the main feature of 2nd law of thermodynamics is that it defines the directions of the processes.

2nd law of thermodynamics defines the process's direction in terms of a system property known as entropy. Thus, entropy is the randomness or disorder of a thermodynamics system. It is an extensive property of a system that is used to measure the amount of energy that is unavailable for work.

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...impossi...
...que It is denoted by S. The equation of change of entropy is
...le th...
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...ess.

$$\Delta S = \frac{Q}{T} \quad \text{—(1)}$$

where, Q = amount of heat energy transfers during a process
T = absolute temp at which the process takes places

we know that, Entropy change is equal to the sum of entropy change of system and surroundings. Therfore.
Total change of entropy

$$(\Delta S)_T = (\Delta S)_{system} + (\Delta S)_{surrounding} \quad \text{—(2)}$$

If the system loses an amount of heat Q at a temp. T_1 which is received by surroundings at temp. T_2.

So, $(\Delta S)_{Total}$ can be calculated

$$(\Delta S)_{system} = -\frac{Q}{T_1} \text{ and } (\Delta S)_{surrounding} = +\frac{Q}{T_2}$$

$$\therefore (\Delta S)_{Total} = -\frac{Q}{T_1} + \frac{Q}{T_2} \quad \text{—(3)}$$

If $(\Delta S)_{Total}$ = +ve, the process is spontaneous
" $(\Delta S)_{Total}$ = -ve, " " " non-spontaneous
" $(\Delta S)_{total}$ = 0, the process is at equilibrium.

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(II) Statements of 2nd law of thermodynamics

(I) Kelvin statement (About heat engine):

" It is impossible to get continuous work from a body by cooling it to a temp. lower than that of its surroundings "

This means that in heat engine, working substance absorbs heat from source and performs mechanical work then rejects certain amount of heat to the sink. But, no any heat engine converts all heat extracted from the source into work without rejecting to the sink. ~~In the~~

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in the other words, presence of colder body is required for ~~other word~~ continuous work. This means single heat reservoir cannot transfer heat into mechanical work continuously.

② Clausius statement (about refrigenerater)

"It is impossible to make flow of heat from colder body to hotter body without doing external work on the working substance".

This means, in a refrigenerater, working substance absorbs heat from the sink, certain external work is added on it so that finally large amount of heat is rejected to the source. But, no one refrigenerter transfer heat from colder body to hotter body without adding external work on it.

Ⅱ Heat engine:

Heat engine is a device which converts heat into mechanical work continuously. For example: the gasoline engines in automobile, airplane, steam turbine of steam engine. are heat engines. Heat engine consists of following components.

Ⅰ Source: It is a body of infinite heat capacity maintained at heat temp. Source has greater temp. (T_1 K) than that of sink. Working substance can absorb any amount of heat from source keeping source temp. constant.

Ⅱ Working substance: The material body of heat engine which burn to produce heat so that it perform mechanical work.

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Ideal gas in carnot's engine, 2% petrol vapour and 98% air in petrol engine are examples of working substance.

(iii) Sink: It is a body of infinite heat capacity maintained at low temp. Sink has lower temp. (T_2 K) than that of source. Any amount of heat can be rejected to sink keeping sink temp. constant.

(iv) Thermal ~~efficiency~~ efficiency of heat engine.

Heat engined operates on a cyclic process and receive heat from a high temp. reservoir (source) convert part of this heat to work and then reject the remaining part of heat to a lower temp. reservoir (sink) during the cycle.

Suppose, working substance extract Q_1 amount of heat from the source at temp. T_1 K, perform certain mechanical work W and rejects remaining amount of heat Q_2 to the sink at lower temp. T_2 K as shown in figure.

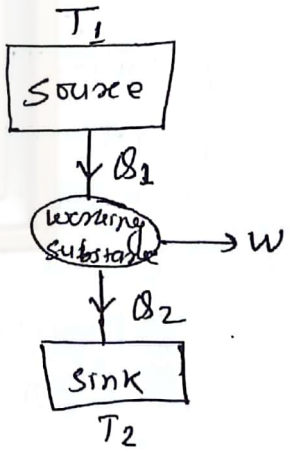


Fig: Block diagram of heat engine

Therefore, $(Q_1 - Q_2)$ is the amount of heat converted into mechanical work. That is work done

$$W = Q_1 - Q_2 \quad \text{—①}$$

Now, efficiency of heat engine is defined as the ratio of the external work obtained to the heat energy absorbed by the working substance from the source. It is denoted by η. So,

$$\eta = \frac{\text{External work done } (W)}{\text{Heat energy absorbed from the source } (Q_1)}$$

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$\therefore \eta = \frac{Q_1 - Q_2}{Q_1} \Rightarrow 1 - \frac{Q_2}{Q_1}$ since $Q_1 - Q_2 = W$.

In general efficiency is expressed in %. So,

$\eta = \left(1 - \frac{Q_2}{Q_1}\right) \times 100\%$ Since $Q_1 > Q_2$, efficiency is always less than 100%. ~~Hundred~~ Per cent percentage efficiency is possible only when $Q_2 = 0$ but $Q = 0$ is impossible because to drive engine, heat must be lost to the sink compulsorily.

In term of temp. efficiency of heat engine can be expressed as

$$\eta = \left(1 - \frac{T_2}{T_1}\right) \times 100\%$$

(ii) Refrigerator:

Refrigerator operates on a cyclic process which takes heat from a low-temp. reservoir at temp. T_2 and delivers into the high-temp. reservoir at temp. T_1 with the help of external work. Refrigerator maintains the temp. of desired space lower than that of surroundings.

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A refrigerator is essentially a heat engine working in the reverse direction. That means in refrigerator, working substance takes heat from cold body (interior of refrigerator), compressor supplies the necessary mechanical work input and sum of two energies is rejected to the hot reservior (surrounding air). The schematic operation of a refrigerator is shown in figure below.

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Suppose working substance absorbs Q_2 amount of heat from sink at temp T_2 on adding external work W unit by motor. Finally, large amount of heat Q_1 is rejected to the source at higher temp. T_2. Now, $Q_1 = Q_2 + W$. —(1)

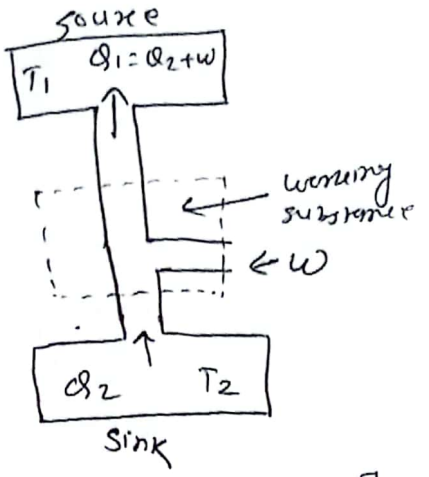


Fig: Block diagram of refrigerator

Here, Input work (or energy) $W = Q_1 - Q_2$

Output work (or energy) $= Q_2$

The efficiency of different refrigerators can be compared by their coefficient of performance (COP) β. Coefficient of perference (β) is defined as the ratio of heat extracted from the cold body (refrigerant) to work done by the motor.

$$\therefore \beta = \frac{\text{Output}}{\text{input}} = \frac{Q_2}{Q_1 - Q_2} \Rightarrow \frac{1}{\frac{Q_1 - Q_2}{Q_2}} \Rightarrow \frac{1}{\left(\frac{Q_1}{Q_2} - 1\right)}$$

$$\therefore \beta = \frac{1}{\left(\frac{T_1}{T_2} - 1\right)} \Rightarrow \beta = \frac{T_2}{T_1 - T_2}$$

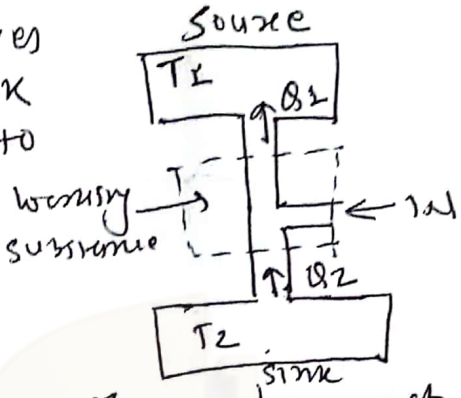
This shows that all reversible refrigerating machines working between the same two temp. have the same coefficient of performance. It is clear from expression, that if $T_1 - T_2$ is small, β will be higher. The best refrigerator removes maximum heat Q_2 with least possible work input by motor. If no work were needed to operate a refrigerator, the coefficient of performance β would become infinite which is rather impossible. For practical refrigerator, β is always less than $\frac{T_2}{T_1 - T_2}$.

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(II) Heat pump

Heat pump operates on a cyclic process which takes heat from a low temp. reservoir (surrounding) at T_2 and delivers into the high temp. reservoir (desired space) at T_1 with the help of external work done. Heat pump maintains the temp. of desired space higher than that of surroundings.

Consider a heat pump takes Q_2 amount of heat from sink at temp. T_2 and supply Q_1 to desired space at T_1 with the help of external work done (W). performance of heat pump is measured in term of coefficient of performance (COP) β, which is defined as the ratio of desired effect and work supplied. i.e.



[Block diagram of heat pump]

$$\beta = \frac{\text{Desire effect}}{\text{work input}}$$

where Desire effect = amount of heat supplied to the desired space (Q_1).

$$\therefore (\beta)_{\text{heat pump}} = \frac{Q_1}{W} = \frac{Q_1}{Q_1 - Q_2} \Rightarrow \frac{1}{1 - \frac{Q_2}{Q_1}}$$

In term of temp,

$$(\beta)_{\text{heat pump}} = \frac{T_1}{T_1 - T_2} \Rightarrow \frac{1}{1 - \frac{T_2}{T_1}}.$$

(III) Components of refrigerator and heat pump:

OR.

Construction and working of Refrigerator.

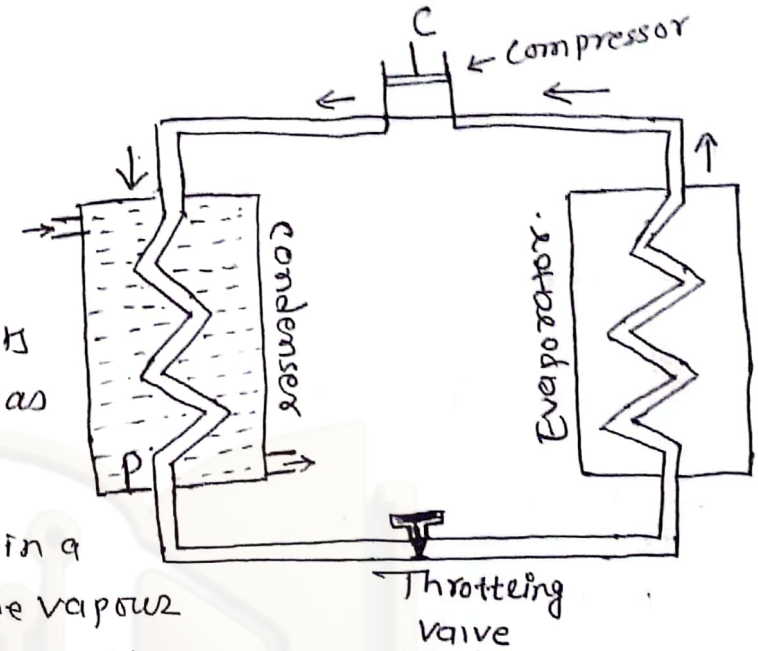
⇒ A refrigerator and heat pump operates on a cyclic process called refrigeration cycle. The refrigeration cycle is the

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widly used for heating and cooling of desired space.

The working principle of common refrigerator has been explained with the support of schematic diagram as shown in figure below. Four components of refrigerator are

(i) Compressor
ii) Condenser
iii) Throttling valve, and
iv) Evaporator.



[Fig: schematic diagram of refrigerator.]

The working procedure of refrigerator along with its components are described as following.

The working substance in a refrigerator is usually the vapour of some low boiling point liquid, such as chloro-fluro carbon (CCl_2F_2) or freon), ammonia, sulphur dioxide etc. The compressor (C) compresses the working substances adiabatically to a high pressure or with the support of motor (external work is done here) so that gas becomes quite hot. This hot and high pressured gas is passed through condenser. condenser is a spiral pipe (P) to which cold water is circulated. Now gas gets liquified. This liquid is passed through the throttling valve (V) and it expands suddenly by the process called throttling process. This process causes the decreases in the temp. of gas. Then the gas is passed through the evaporater (spiral tube → covering department containing objects to be cooled like food items) where

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liquid evaporates and gets cooled by taking heat from food items. Now, working substance leaves the cooling unit and then goes to the compressor (C) for fresh cycle.

Note:

Compressor → high pressure - Temp. → P, T and H increases but entropy(S) remain cent.

Condenser → high pressure - low temp. → P = const. H and S decreases.

Throttling valve → high press - low temp → H = const, P & T = decreases and S = increases.

Evaporater → low pressure - low temp → P = const, H & S = increases.

(II) Numerical problems:

① A gas is contained in a piston cylinder device initially at a pressure of 150 kPa and a volume of 0.04 m^3. Calculate the work done by the gas when it undergoes the following process to a final volume of 0.1 m^3.

(a) constant pressure. (b) Constant temp. (c) $PV^{1.35}$ = constant.

Soln: we have given,

initial pressure (P_1) = 150 kPa

" Volume (V_1) = 0.04 m^3

Final " (V_2) = 0.1 m^3.

work done by the gas = ?

(a) At Const. pressure, work done $W = \int pdV = P_1(V_2 - V_1)$

$\therefore W = 150 \times 10^3 (0.1 - 0.04) \Rightarrow W = 9000\ J.$

(b) At constant temp., work done $W = P_1V_1 \ln(V_2/V_1)$

$= 150 \times 10^3 \times 0.04 \ln\left(\frac{0.1}{0.04}\right) = 5497.7\ J.$

(c) For adiabatic process. PV^γ = const. Since $\gamma = 1.35$.

$P_1V_1^\gamma = P_2V_2^\gamma \Rightarrow P_2 = P_1\left(\frac{V_2}{V_1}\right)^\gamma = 1.35$

$P_2 = 150 \times 10^3 \left(\frac{0.04}{0.1}\right)^{1.35} \Rightarrow 43538.38\ Pa.$

$\frac{1}{1-1.35}$ (Used the value at all then calculate find

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(2) An ideal engine has an efficiency of 25%. If the source temp. is increased by 200°C, its efficiency gets doubled. Determine its source and sink temp.

Soln.

Let Temp. of source = T_1

" " sink = T_2

efficiency $\eta = 1 - \frac{T_2}{T_1} \Rightarrow 0.25 = 1 - \frac{T_2}{T_1} \Rightarrow T_2 = 0.75 T_1$. — (1)

η is double, i.e. $\eta_2 = 2\eta = 2 \times 0.25 \Rightarrow \eta_2 = 0.50$.

$\therefore \eta_2 = 1 - \frac{T_2}{T_1}$

$0.50 = 1 - \frac{T_2}{T_1 + 200}$

$\frac{T_2}{T_1 + 200} = 0.5$ — (2)

Using eq (1) in eq (2). We get

$\frac{0.75 T_1}{T_1 + 200} = 0.5$

$\therefore T_1 = 400\ K$.

and $T_2 = 0.75 \times 400 = 300\ K$ //

(3) The source reservoir of a heat engine is at temp. of 400 K and takes 400 J of heat and rejects 20 J of heat to the sink reservoir in each cycle. What are the efficiency of engine and temp. of sink?

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Soln Hints: $T_1 = 400\ K$, $T_2 = ?$ $Q_1 = 400\ J$, $Q_2 = 20\ J$ $\eta = ?$

$\eta\% = \frac{Q_1 - Q_2}{Q_1} \times 100\% = \frac{400 - 20}{400} \times 100\% = 95\%$.

Again. $\frac{Q_1}{Q_2} = \frac{T_1}{T_2} \Rightarrow \frac{400}{20} = \frac{400}{T_2} \Rightarrow T_2 = 20\ K$.

(4) An ideal heat engine operates between two reservoir at two temp in order to achieve 30% efficiency when the temp. of the sink is 50°C. What should be the temp. of the source?

Soln. Hints $\eta = 30\%$ $T_2 = 50°C = (50 + 273)\ K = 323\ K$

$T_1 = ?$ We have $\frac{T_1 - T_2}{T_1} \times 100\% = \eta\% \Rightarrow T_1 = 461.43\ K$.

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Electromagnetic wave

(I) Vector algebra:

$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$

$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$

(i) Dot product: $\vec{A}\cdot\vec{B} = (A_x\hat{i} + A_y\hat{j} + k_z\hat{k})\cdot(B_x\hat{i} + B_y\hat{j} + B_z\hat{k})$

$= A_xB_x + A_yB_y + A_zB_z$

(ii) Vector product: $\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \hat{i}(A_yB_z - B_yA_z) + \hat{j}(A_zB_x - B_zA_x) + \hat{k}(A_xB_y - B_xA_y)$

(II) Vector operator:

$\nabla = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$ This quantity gives the variation of the various quantities quantitatively along different direction.

(III) New terms:

(i) Gradient of a scalar
(ii) Divergence of a vector
(iii) Curl of a vector.

} They are described as following

(i) Gradient of a scalar:

Let ϕ be a scalar function, then gradient of ϕ is given by $\nabla\phi$ and is given as $\nabla\phi = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\phi$

$$\nabla\phi = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}$$

Hence, Gradient of a scalar is a vector quantity.

Physical significance of gradient:

Gradient of a scalar is maximum slope at point. The change of vector with respect to ~~vector~~ position is known as gradient physically. Gradient gives the magnitude and direction of maximum change of the quantity along the particular direction.

(ii) Divergence of a vector:

Let $\vec{A}$ be a vector quantity. Its divergence is denoted by $\nabla\cdot\vec{A}$ and is given by $\nabla\cdot\vec{A} = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\cdot(\hat{i}A_x + \hat{j}A_y + \hat{k}A_z)$

$$\nabla\cdot\vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$ which is scalar

Hence, divergence of a vector quantity is a scalar quantity.

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Physical significance of Divergence:

Divergence of any vector quantity gives scalar. There are Three cases. They are

(i) Case-1: Divergence of vector is +ve
i.e. $\nabla \cdot \vec{A} > 0$ it means, it is a source, i.e. flow of flux will be increased

(ii) Case-II: Divergence of vector is -ve,
i.e. $\nabla \cdot \vec{A} < 0$, it means it is a sink, i.e. flow of flux will be decreased

(iii) Case-III: Divergence of vector is zero
i.e. $\nabla \cdot \vec{A} = 0$, it means, there is ~~neither~~ flow of flux neither increase nor decrease, and going similar out of given input.

(iv) Curl of a vector:

Let $\vec{A}$ be a vector quantity, then curl of $\vec{A}$ is denoted by

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \hat{i}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{j}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{k}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

Physical significance of curl of a vector:

Curl of any vector gives the rotation of the quantity.

(II) Gauss divergence theorem:

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It states that the volume integral of divergence of a vector $\vec{A}$ is equal to the surface integral of $\vec{A}$ takes over the surface which encloses the volume.

$$\therefore \oint_V (\nabla \cdot \vec{A})\, dv = \oint_S \vec{A} \cdot \hat{n}\, ds$$

(II) Stoke's Theorem:

It states that the surface integral of a curl of a vector $\vec{A}$ takes over the surface is equal to the line integral of vector $\vec{A}$ taken over peripheral of $\vec{A}$.

$$\text{i.e. } \oint_S (\nabla \times \vec{A}) \cdot d\vec{s} = \oint_l \vec{A} \cdot d\vec{l}$$

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(H) Continuity equation [imp]

Current is given by $I = -\frac{dq}{dt}$ —① where -ve sign shows that charge decreases with time. Since $\oint I = \oint_S \vec{J} \cdot d\vec{s}$ where $\vec{J}$ is the electric current density.

Since $\vec{J} = \frac{\vec{I}}{A} \Rightarrow I = \vec{J} \cdot \vec{A} = \oint_S \vec{J} \cdot d\vec{s}$

Also, $q = \oint_V \rho \, dv$ since $\rho = \frac{q}{v} \Rightarrow q = \rho \cdot v = \oint \rho \, dv$

Equation ① becomes

$$I = \oint_S \vec{J} \cdot d\vec{s} = -\frac{d}{dt} \oint_V \rho \, dv \quad —②$$

Using gauss divergence theorem.

$$\oint_S \vec{J} \cdot d\vec{s} = \oint_V (\nabla \cdot \vec{J}) \, dv$$

So, eq^n ② will be

$$\oint_V (\nabla \cdot \vec{J}) \, dv = -\frac{d}{dt} \oint_V \rho \, dv$$

$$\text{or } \oint_V (\nabla \cdot \vec{J}) \, dv = -\oint_V \left(\frac{d\rho}{dt}\right) dv$$

$$\text{or } \oint_V (\nabla \cdot \vec{J}) \, dv = \oint_V \left(-\frac{\partial \rho}{\partial t}\right) dv$$

$$\text{or } \nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\therefore \nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0 \quad —③$$

This is the equation of continuity which explains the law of conservation of charge.

For the steady state condition, i.e., for static charge,

$\frac{\partial \rho}{\partial t} = 0$ so, eq^n ③ will be $\nabla \cdot \vec{J} = 0$, Thus divergence of current will be zero means, there is no source and sink.

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